

Software Requirements Specification (SRS)

For

Nexora: Multi-Vendor E-Commerce Marketplace

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Prepared By: Enterprise Architecture & Product Engineering Group

2. Document Information

Attribute	Details
Project Title	Nexora Multi-Vendor E-Commerce Marketplace
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Target Audience	Enterprise Architecture Boards, Engineering Leads, Product Stakeholders, QA Leads, DevOps Leads, Client Evaluation Committee
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		Architect & Principal BA	enterprise SRS generation for production engineering kickoff.
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5. Introduction

5.1 Purpose

This Software Requirements Specification (SRS) defines the complete functional, non-functional, operational, and governance requirements for **Nexora**, an enterprise-grade multi-vendor e-commerce marketplace. The document serves as the formal binding contract between product engineering teams, stakeholders, and quality assurance divisions, dictating the precise behavioral specifications of the platform prior to development commencement.

5.2 Document Conventions

- **SHALL** or **MUST**: Indicates a mandatory requirement that must be implemented without exception.
- **SHOULD**: Indicates a recommended feature or behavior that is highly desirable unless technical constraints dictate otherwise.
- **MAY**: Indicates an optional or secondary capability.
- Industry-standard terminology regarding e-commerce, distributed caching, event-driven architecture, and role-based access control applies throughout.

5.3 Intended Audience and Reading Suggestions

- **Executive Leadership**: Focus on Sections 6 through 10 (Vision, Objectives, Scope).
- **Product Managers & Business Analysts**: Focus on Sections 13 through 16 (Permissions, Functional Specs, Business Rules).
- **Software Architects & DevOps Engineers**: Focus on Sections 15, 18, 19, and 20 (NFRs, Constraints, System Modules, Workflows).
- **Quality Assurance Engineers**: Focus on Sections 14, 16, and 21 (Functional Modules, Business Rules, Use Cases).

6. Project Overview

Nexora is designed as a highly scalable, multi-tenant-capable multi-vendor e-commerce marketplace. It bridges independent merchants (sellers) and digital consumers through an intuitive, high-performance web interface backed by a robust enterprise server architecture. The platform acts as a trusted intermediary, managing catalog distribution, secure financial transaction processing, order routing, multi-tier fulfillment tracking, automated commission calculations, and comprehensive dispute resolution workflows.

7. Business Problem

Traditional single-vendor online stores limit product diversity, fail to provide localized shipping competition, and lack scalable inventory distribution models. Conversely, legacy multi-vendor marketplaces often suffer from:

- **Monolithic Bottlenecks**: System crashes during peak shopping events due to tight component coupling.
- **Inventory Desynchronization**: Over-selling products because of high-latency inventory tracking across disparate vendor stocks.
- **Opaque Settlement Cycles**: Manual, error-prone financial payouts to third-party vendors resulting in compliance and trust deficits.
- **Security and Authorization Flaws**: Inadequate separation of duties allowing unauthorized catalog manipulation or data breaches.

Nexora solves these challenges by establishing an isolated, secure, event-driven, modular platform built on strict clean-architecture principles.

8. Project Vision

To establish the premier digital commerce ecosystem recognized for uncompromising

transactional integrity, lightning-fast product discovery, transparent merchant collaboration, and seamless horizontal scalability capable of handling enterprise-grade traffic volumes without degradation.

9. Project Objectives

- **High Availability:** Maintain 99.99% uptime availability during standard and peak retail operations.
- **Rapid Onboarding:** Enable verified external sellers to configure their storefronts, upload inventories, and initiate sales within 24 hours of registration.
- **Transactional Safety:** Ensure atomic operations for order placements, payment settlements, and inventory decrements to prevent race conditions.
- **Extensibility:** Design core backend services using decoupled modular boundaries to support future migration to containerized microservices.
- **Optimized Performance:** Deliver sub-second catalog search responses and sub-200ms page transition latencies via caching and efficient database indexing.

10. Project Scope

In-Scope

- Multi-vendor registration, vetting, verification, and role-based portal access.
- Dynamic category management and hierarchical attribute-driven product catalogs.
- Real-time inventory tracking, warehouse management, and low-stock alerts.
- Customer cart, wishlist management, multi-shipping-address checkout, and secure payment processing.
- Order lifecycle execution (Placement, Authorization, Fulfillment, Delivery, Cancellation, Refund).
- Automated vendor commission splits and payout ledgers.
- Ratings, verified purchase reviews, moderation tools, and dispute logging.
- Comprehensive administrative reporting, audit logs, notification engines, and analytics dashboards.

Out-of-Scope

- Native mobile applications (iOS/Android) for initial release (responsive web interface is supported).
- Physical logistics fleet management (handled via integrated third-party shipping provider APIs).
- Cryptocurrency or offline barter payment mechanisms (restricted to standard fiat gateways and credit instruments).

11. Stakeholders

Stakeholder Group	Role & Responsibilities
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Platform Administrators	Govern platform rules, review seller applications, handle dispute escalations, and monitor system metrics.
Independent Sellers	Manage digital store catalogs, fulfill orders, configure shipping rules, and track revenue disbursements.
End Customers	Discover products, execute secure purchases, manage accounts, and submit reviews.
Engineering & DevOps Team	Develop, test, secure, deploy, and maintain the software ecosystem across environments.
Payment Gateway Providers	External financial institutions processing credit cards, digital wallets, and net banking transactions.
Logistics Partners	External courier services delivering packages and providing real-time tracking webhooks.

12. Target Users

- **B2C Retail Shoppers:** Individuals seeking diverse product choices, secure checkout experiences, and reliable order tracking.
- **Small-to-Medium Business (SMB) Merchants:** Independent brand owners and wholesale distributors looking for a ready-made digital storefront to reach wider consumer bases.
- **Platform Operations Managers:** Internal corporate staff responsible for auditing transactions, resolving vendor-customer conflicts, and maintaining category taxonomies.

13. User Roles and Permissions Matrix

Feature / Module	Guest	Customer	Seller	Administrator
Browse Public	Yes	Yes	Yes	Yes

Catalog & Search				
Register Account / Login	Yes	Yes	Yes	Yes
Manage Personal Profile / Password	No	Yes	Yes	Yes
Manage Shopping Cart & Wishlist	No	Yes	No	No
Place & Track Personal Orders	No	Yes	No	No
Submit Product Reviews & Ratings	No	Yes (Verified Only)	No	Yes (Moderate)
Apply for Seller Account	No	Yes	No	No
Manage Own Storefront & Profile	No	No	Yes	Yes
Manage Own Product Catalog & Pricing	No	No	Yes	Yes

Manage Own Inventory & Stock Levels	No	No	Yes	Yes
Fulfill & Update Status of Assigned Orders	No	No	Yes	Yes
View Own Sales Analytics & Reports	No	No	Yes	Yes
Approve / Reject Seller Applications	No	No	No	Yes
Global User & Role Administration	No	No	No	Yes
Global Category & Taxonomy Management	No	No	No	Yes
System-Wide Reports & Financial Audits	No	No	No	Yes
Platform Configuration & Feature Toggles	No	No	No	Yes

14. Functional Requirements

14.1 Authentication Module

- The system shall allow users to register with a unique email address and a secure password.
- The system shall issue JSON Web Tokens (JWT) upon successful authentication to secure subsequent API calls.
- The system shall support secure password reset flows via email token verification.
- The system shall enforce role-based access control (RBAC) ensuring endpoints restrict access based on assigned user roles.
- The system shall invalidate active tokens upon user logout.

14.2 User Management Module

- The system shall allow users to update their profile information including first name, last name, phone number, and avatar.
- The system shall allow customers to manage multiple shipping and billing addresses.
- The system shall allow administrators to view, search, activate, or deactivate any user account.
- The system shall maintain an immutable audit trail of user account status modifications.

14.3 Seller Management Module

- The system shall allow registered users to submit a seller onboarding application containing business details, tax identification numbers, and payout bank information.
- The system shall allow administrators to review, approve, or reject seller applications with mandatory rejection justifications.
- The system shall allow approved sellers to customize their public storefront profile banner, logo, and store description.
- The system shall suspend seller capabilities automatically if regulatory compliance checks or subscription validations fail.

14.4 Product Management Module

- The system shall allow sellers to create, read, update, and soft-delete products within their assigned inventory.
- The system shall support multiple product types including simple products and configurable products with distinct variants (e.g., size, color).
- The system shall allow uploading and management of multiple high-resolution product images per item.
- The system shall enforce mandatory attributes for products based on their assigned category taxonomy.

14.5 Category Management Module

- The system shall allow administrators to create a hierarchical category tree with parent-child relationships.
- The system shall allow assigning products to leaf categories within the hierarchy.
- The system shall allow administrators to reorder, enable, disable, or delete categories.

14.6 Inventory Module

- The system shall track real-time stock quantities for every individual product variant.
- The system shall automatically decrement stock quantities upon successful order payment confirmation.
- The system shall release reserved stock back into inventory if an initiated checkout transaction expires or fails.
- The system shall trigger automated low-stock alerts to sellers when stock falls below configured threshold levels.

14.7 Cart Module

- The system shall allow authenticated customers to add, update quantity, or remove items from their persistent shopping cart.
- The system shall validate product availability and current pricing whenever a customer accesses their shopping cart.
- The system shall merge guest cart contents with user account carts upon successful login.

14.8 Wishlist Module

- The system shall allow customers to maintain a personal wishlist of desired products.
- The system shall allow customers to move items directly from their wishlist into their active shopping cart.
- The system shall notify customers if a wishlisted item goes out of stock or undergoes a price reduction.

14.9 Order Module

- The system shall allow customers to convert their shopping cart into an official order by selecting delivery addresses and payment methods.
- The system shall split multi-vendor cart checkouts into distinct sub-orders assigned to individual sellers.
- The system shall track order lifecycles through states: PENDING, PAYMENT_CONFIRMED, PROCESSING, SHIPPED, DELIVERED, CANCELLED, and REFUNDED.
- The system shall allow sellers to update sub-order fulfillment statuses and attach shipment tracking numbers.

14.10 Payment Module

- The system shall integrate with secure third-party payment gateways to process credit card, debit card, and digital wallet transactions.
- The system shall handle asynchronous payment webhook callbacks to confirm transaction success or failure securely.
- The system shall record detailed payment transaction logs including gateway response codes and timestamps.
- The system shall support full and partial refund processing linked to approved order return requests.

14.11 Shipping Module

- The system shall allow sellers to define custom shipping rates or integrate with regional courier calculation engines.
- The system shall calculate accurate shipping fees during checkout based on delivery destination, package weight, and dimensions.
- The system shall capture and display real-time shipment milestone tracking updates to customers and administrators.

14.12 Reviews Module

- The system shall allow verified purchasers to submit star ratings (1 to 5) and textual reviews for delivered products.
- The system shall allow sellers to reply to customer reviews in a public forum.
- The system shall allow administrators to flag and remove inappropriate, abusive, or spam reviews.

14.13 Notifications Module

- The system shall dispatch automated transactional emails and system notifications for key events (registration, order placement, shipping updates, password resets).
- The system shall maintain an in-app notification center for users to view unread alerts.

14.14 Dashboard Module

- The system shall provide role-specific dashboard interfaces summarizing critical metrics (sales volume for sellers, pending tasks for administrators, recent orders for customers).
- The system shall render interactive charts representing sales trends over selectable time intervals.

14.15 Analytics Module

- The system shall aggregate sales performance data, conversion rates, and product popularity metrics for administrative and seller review.
- The system shall compute top-selling categories and geographic purchasing distributions.

14.16 Reports Module

- The system shall allow administrators and sellers to export financial sales reports, tax summaries, and inventory logs in CSV and PDF formats.
- The system shall schedule automated generation and distribution of periodic business reports.

14.17 Search Module

- The system shall provide full-text search capabilities across product titles, descriptions, and brand attributes.
- The system shall support search auto-completion and spell-correction suggestions.

14.18 Filtering Module

- The system shall allow users to refine search results by price range, category, brand, customer rating, and dynamic attribute filters.
- The system shall update result counts dynamically as filters are applied.

14.19 Pagination Module

- The system shall paginate all list views, search results, order histories, and administrative tables with configurable page sizes.
- The system shall provide navigation controls for traversing result sets efficiently.

15. Non-Functional Requirements

15.1 Performance

- Page load times for catalog browsing shall not exceed 1.5 seconds under normal operational load.
- REST API response times for read operations shall remain under 200 milliseconds at the 95th percentile.
- The system shall support concurrent user sessions up to 10,000 active connections without performance degradation.

15.2 Security

- All data transmitted between clients and servers shall be encrypted using TLS 1.3.
- Passwords shall be salted and hashed using industry-standard cryptographic algorithms (e.g., bcrypt with a work factor of 12).
- The system shall protect against common web vulnerabilities including SQL Injection, Cross-Site Scripting (XSS), Cross-Site Request Forgery (CSRF), and Parameter Tampering.
- Role-based authorization checks shall be enforced at every API boundary layer.

15.3 Scalability

- The architecture shall be horizontally scalable, allowing addition of compute node instances behind load balancers.

- Database query execution shall be optimized via strategic indexing, connection pooling, and distributed caching layers (Redis) for high-frequency read operations.

15.4 Availability

- The platform shall maintain an uptime availability of 99.99%, allowing no more than 52 minutes of unscheduled downtime annually.
- Critical components shall be deployed with redundant failover configurations.

15.5 Reliability

- Transactional operations affecting inventory and payments must adhere to ACID properties.
- Asynchronous message brokers (Kafka) shall guarantee at-least-once delivery semantics for event-driven workflows.

15.6 Logging

- Comprehensive structured logs (JSON format) shall be captured for all authentication attempts, financial transactions, administrative actions, and critical system errors.
- Log levels (DEBUG, INFO, WARN, ERROR, FATAL) shall be dynamically configurable without application restarts.

15.7 Exception Handling

- The system shall return standardized, user-friendly error response payloads containing unambiguous error codes, error descriptions, and timestamps for all client-facing failures.
- Unhandled runtime exceptions must be securely caught, logged with stack traces internally, and masked from external clients.

15.8 Validation

- All incoming request payloads shall undergo strict structural and semantic validation before business logic execution.
- Validation failure responses shall detail specific field-level violations to assist client integration.

15.9 Maintainability

- The codebase shall adhere to strict modular layered architectures, promoting separation of concerns and adherence to SOLID design principles.
- Unit test coverage across business logic modules shall be maintained at a minimum of 80%.

15.10 Deployment

- The application shall be fully containerized using Docker to guarantee environment parity across development, staging, and production clusters.

- Continuous Integration and Continuous Deployment (CI/CD) pipelines shall automate building, testing, and deployment workflows.

15.11 Monitoring

- System health check endpoints shall expose vital health indicators (database connectivity, disk space, memory utilization, cache availability) for automated infrastructure monitoring tools.
- Real-time metrics collection shall track CPU usage, memory consumption, request throughput, and error rates.

15.12 Backup

- Automated incremental database backups shall be executed every 6 hours, with full daily backups retained in geographically redundant cloud storage for 30 days.
- Disaster recovery restore procedures shall be tested and verified on a quarterly basis.

16. Business Rules

1. A user must verify their email address before they can activate their customer or seller account.
2. A single email address can only be associated with one active user account in the system.
3. Passwords must be at least 8 characters long and contain at least one uppercase letter, one lowercase letter, one number, and one special character.
4. An unauthenticated guest user may browse products, categories, and reviews, but cannot add items to a cart or execute purchases.
5. A customer account can have a maximum of 10 saved shipping addresses at any given time.
6. A seller application must include a valid tax identification number and verified corporate banking details before submission.
7. Administrators must review and approve or reject seller applications within 5 business days of submission.
8. If a seller application is rejected, the administrator must provide a written reason that is displayed to the applicant.
9. A suspended seller cannot create new products, edit existing products, or fulfill new orders.
10. A product SKU (Stock Keeping Unit) must be globally unique across the entire platform.
11. A product price must be a positive decimal value greater than zero.
12. A product promotional discount price cannot exceed the original listed product price.
13. A seller can only assign products to active leaf categories within the category hierarchy.
14. Product inventory quantities cannot be negative under any circumstance.
15. When a customer adds an item to their cart, stock availability is checked but not permanently reserved until checkout initiation.
16. Items placed in a customer's shopping cart expire and are removed automatically after 7 days of inactivity.

17. Items placed in a customer's wishlist remain indefinitely until manually removed or purchased.
18. During checkout, if multiple items belong to different sellers, the system must automatically split the cart into separate sub-orders per seller.
19. Shipping fees are calculated independently for each sub-order based on the respective seller's shipping rule configurations.
20. A customer has a maximum of 15 minutes to complete payment once an order is placed before the order status changes to EXPIRED and reserved stock is released.
21. Payment gateway webhook notifications must be cryptographically verified using shared secrets before processing.
22. An order status cannot transition backwards (e.g., an order marked as DELIVERED cannot be changed back to PROCESSING).
23. A customer may request an order cancellation only if the sub-order status is PENDING or PAYMENT_CONFIRMED.
24. Once a sub-order status reaches SHIPPED, cancellation is no longer permitted; the customer must initiate a return request post-delivery.
25. Sellers must input a valid tracking number from an approved courier partner when marking a sub-order as SHIPPED.
26. A customer can submit a product review only if their order containing that product has reached DELIVERED status.
27. A customer can submit only one review per purchased product variant.
28. Administrators hold the right to moderate, hide, or permanently delete any customer review that violates community guidelines.
29. Sellers may reply to customer reviews once per review, and replies are subject to administrative moderation.
30. Platform commission is automatically deducted from the seller's total order subtotal upon successful payment confirmation.
31. The standard platform commission rate is set at 10% but can be overridden per individual seller agreement by administrators.
32. Seller payout settlements are calculated weekly and disbursed only after a 14-day holding period to accommodate potential customer returns and refunds.
33. Refund amounts issued to customers must include proportional shipping fee refunds if an entire sub-order is returned.
34. Partial refunds issued by sellers must not exceed the total item value of the specific sub-order.
35. Low-stock notifications are dispatched to sellers when product variant inventory drops to 5 units or fewer.
36. Out-of-stock products are automatically hidden from search results or marked as "Out of Stock" based on administrator configuration settings.
37. A user account is automatically locked for 30 minutes after 5 consecutive failed login attempts due to incorrect passwords.
38. Administrators can manually unlock locked user accounts upon identity verification.
39. Inactive user accounts showing no login activity for 2 years are automatically archived by

the system.

40. All administrative actions (approvals, deletions, configuration updates) must be recorded in the immutable audit log table.
41. Search queries containing fewer than 2 characters shall be rejected with a validation error.
42. Product filter selections must persist across pagination page transitions within the same browsing session.
43. Currency across all platform transactions is standardized to USD unless multi-currency localization modules are explicitly enabled.
44. Tax calculations during checkout are determined based on the customer's shipping destination address jurisdiction.
45. Sellers cannot purchase their own products using their seller account credentials.
46. Promotional coupon codes can only be applied once per customer per promotional campaign unless specified otherwise.
47. Expired coupon codes are automatically rejected during the checkout calculation phase.
48. File uploads for product images are restricted to JPEG, PNG, and WebP formats, with a maximum file size limit of 5 megabytes per image.
49. Storefront banners uploaded by sellers must adhere to a minimum resolution of 1200x400 pixels.
50. System maintenance mode can only be activated by users with Super Administrator privileges, suspending all non-admin API traffic with a 503 Service Unavailable response.

17. Assumptions

- Users possess modern web browsers with JavaScript enabled and stable internet connectivity.
- Third-party payment gateways and courier shipping APIs will maintain 99.9% uptime and provide stable webhook delivery mechanisms.
- Sellers are legally compliant entities responsible for their own tax filings and regulatory adherence within their respective jurisdictions.
- Initial product catalog data migration (if applicable) will be provided in standardized CSV format.
- Cloud infrastructure providers will supply reliable persistent storage and database instances matching performance specifications.

18. Constraints

- The backend software must be developed using Java and Spring Boot, utilizing Maven for dependency management.
- Relational data storage must reside exclusively in MySQL, while distributed caching requires Redis.
- Event-driven messaging must be implemented via Apache Kafka.
- Frontend interfaces must be engineered using React.
- Source code version control must be maintained strictly through Git and GitHub repositories.

- The system must operate within defined cloud server memory and CPU quotas without exceeding allocated infrastructure budgets.

19. System Modules

The Nexora platform is structured into the following distinct functional modules:

- **Core Kernel Module:** Manages global configuration, security filters, JWT generation, exception handling, and auditing frameworks.
- **Identity & Access Module:** Handles user registration, authentication, RBAC authorization, and password management.
- **User & Profile Module:** Manages customer profiles, address books, and administrative user controls.
- **Seller & Storefront Module:** Governs seller onboarding applications, verification workflows, and storefront customizations.
- **Catalog & Category Module:** Manages hierarchical category trees, product attributes, product variants, and media assets.
- **Inventory & Stock Module:** Tracks real-time stock levels, warehouse allocations, and low-stock triggers.
- **Shopping & Wishlist Module:** Manages persistent shopping carts, cart item validation, and customer wishlists.
- **Order & Fulfillment Module:** Handles order placement, multi-vendor cart splitting, state transitions, and shipment tracking.
- **Payment & Finance Module:** Integrates payment gateways, webhook processing, ledger tracking, commission splits, and payouts.
- **Shipping & Logistics Module:** Calculates freight charges and manages carrier integration hooks.
- **Feedback & Moderation Module:** Manages customer reviews, star ratings, seller replies, and content moderation.
- **Notification & Communication Module:** Dispatches transactional emails, SMS, and in-app alerts.
- **Reporting & Analytics Module:** Aggregates business intelligence, generates sales reports, and renders dashboard metrics.

20. High-Level Workflow

1. **Customer Discovery:** A customer navigates the React frontend, searches for products using filters, and views detailed product specifications.
2. **Cart Management:** The customer adds desired items to their shopping cart and proceeds to checkout.
3. **Checkout & Splitting:** The system validates inventory, calculates taxes and shipping costs, and splits the cart into distinct sub-orders per vendor.
4. **Payment Processing:** The customer selects a payment method and authorizes payment via the secure payment gateway.
5. **Order Confirmation & Event Publishing:** Upon payment success webhook receipt, the system confirms orders, reserves funds, decrements inventory stock, and publishes an

order-created event to Kafka.

6. **Fulfillment:** Respective sellers receive notifications, prepare packages, generate tracking numbers, and update sub-order status to SHIPPED.
7. **Delivery & Review:** The courier delivers the package, the status updates to DELIVERED, and the customer is invited to submit a product review.
8. **Settlement:** Following the 14-day holding period, the platform calculates commissions and schedules vendor payout disbursements.

21. Use Case List

Use Case ID	Use Case Name	Primary Actor	Description
UC-01	Register New User	Guest	Register an account as a customer or prospective seller.
UC-02	Authenticate User	Guest / User	Log in using credentials to receive a secure JWT token.
UC-03	Browse & Search Catalog	Guest / Customer	Search and filter products across categories.
UC-04	Manage Shopping Cart	Customer	Add, update, or remove items in the shopping cart.
UC-05	Execute Checkout	Customer	Place an order and process secure payment.
UC-06	Submit Seller Application	Customer	Apply to become an approved marketplace merchant.
UC-07	Review Seller Application	Administrator	Approve or reject pending seller applications.

UC-08	Manage Product Catalog	Seller	Create, update, or remove products and variants.
UC-09	Update Inventory Levels	Seller	Adjust stock quantities for product variants.
UC-10	Fulfill Sub-Order	Seller	Update order status and add shipping tracking numbers.
UC-11	Submit Product Review	Customer	Rate and review a purchased and delivered product.
UC-12	Moderate Reviews	Administrator	Flag or delete inappropriate customer reviews.
UC-13	Generate Sales Report	Administrator / Seller	Export financial and sales performance reports.
UC-14	Manage Category Tree	Administrator	Create, edit, and organize product categories.
UC-15	Configure System Settings	Administrator	Modify global platform rules and commission rates.

22. Future Enhancements

1. Integration of native mobile applications for iOS and Android platforms.
2. Implementation of real-time multi-vendor live chat support for customer inquiries.
3. Integration of AI-powered personalized product recommendations based on browsing history.
4. Support for multi-currency transactions with automated exchange rate synchronization.

5. Implementation of subscription-based recurring product delivery models (Auto-Ship).
6. Integration of advanced fraud detection machine learning models during checkout.
7. Support for multi-warehouse inventory distribution and intelligent order routing.
8. Implementation of wholesale B2B tiered pricing and bulk quotation negotiation workflows.
9. Integration of social commerce features allowing direct purchasing from social media feeds.
10. Support for gift card issuance, redemption, and digital wallet balance storage.
11. Implementation of automated tax calculation engines using third-party tax compliance APIs.
12. Addition of vendor-specific promotional flash sales and coupon generation tools.
13. Implementation of progressive web application (PWA) offline browsing capabilities.
14. Integration of voice-activated product search and navigation features.
15. Support for auction-style bidding on specialty product listings.
16. Implementation of automated return label generation and reverse logistics tracking.
17. Integration of advanced multi-vendor analytics dashboards with predictive sales forecasting.
18. Support for digital downloadable products and software license key delivery.
19. Implementation of customer loyalty reward points and tiered membership programs.
20. Integration of multi-factor authentication (MFA) via TOTP authenticator apps.
21. Support for decentralized supplier dropshipping integrations via EDI standards.
22. Implementation of automated sentiment analysis on customer product reviews.
23. Addition of localized multilingual storefront translations and right-to-left layout support.
24. Integration of augmented reality (AR) product preview tools for furniture and apparel.
25. Support for collaborative shopping carts shared among multiple users.
26. Implementation of automated vendor performance scoring and tier badge assignments.
27. Integration of carbon-neutral shipping offset calculation options during checkout.
28. Support for headless commerce API integrations for third-party storefront clients.
29. Implementation of automated web accessibility (WCAG 2.1 AA) auditing tools.
30. Migration of modular monolithic backend architecture into fully containerized microservices orchestrated via Kubernetes.

23. Glossary

- **ACID:** Atomicity, Consistency, Isolation, Durability; set of properties that guarantee reliable database transactions.
- **API:** Application Programming Interface; set of definitions and protocols for building and integrating application software.
- **B2C:** Business-to-Consumer commerce model.
- **JWT:** JSON Web Token; compact, URL-safe means of representing claims to be transferred between two parties.
- **RBAC:** Role-Based Access Control; method of restricting system access to authorized users based on assigned roles.
- **SKU:** Stock Keeping Unit; distinct type of item for sale such as a product and all attributes

combined.

- **SSL/TLS:** Secure Sockets Layer / Transport Layer Security; cryptographic protocols designed to provide communications security over a computer network.
- **Webhook:** HTTP callback; a way for an app to provide other applications with real-time information.

24. Appendix

Appendix A: Reference Standards

- OpenAPI Specification 3.0 for REST API documentation.
- OWASP Top 10 Web Application Security Risks mitigation guidelines.
- RESTful API Design Best Practices (Richardson Maturity Model Level 3).

Appendix B: Document Approval Sign-Off

- **Lead Enterprise Architect:** _____ Date: _____
- **Principal Product Manager:** _____ Date: _____
- **Head of Quality Assurance:** _____ Date: _____